

ESRs ASSESSMENT

According to PED's Essential Safety Requirements (Annex I)

Scope of this document is to address how compliance with ESRs has been obtained when non-harmonized code has been adopted.

Remark : Only sentences which follow crossed boxes are applicable.

Product Description: Job or DWG No:	Oil/Gas Separator 20980145-083-001 Rev.3
Company Name:	Well-Tanker (Anhui) M&E Equipment Co.,Ltd.
Company Address:	Wushan Town Hefei City Anhui Province China
Items Art. 4 (°) :	EQ. : ☒ Vessels Par. 1(a)/ Steam generator Par. 1.2 /

For ☒ **vessels**, the Manufacturer has adopted all mandatory provisions of Code:

- ☒ ASME VIII Div. 1 (edition and addenda specified in the drawing), which cover provisions of PED at extent shown on the " Guide for ASME stamp holders" to be used to meet PED.
- ☐ ASME VIII Div. 2 (edition and addenda specified in the drawing).
- ☐ GB150-2011 or GB151-1999
- ☐ ISPESL VSR / S / M Rev. 1995 Ed. 1999.
- ☐ PD 5500 (edition and addenda specified in the drawing).

For ☐ **steam generators**, the Manufacturer has adopted all mandatory provisions of Code:

- ☐ ASME I (edition and addenda specified in the drawing).
- ☐ ASME B31.1 (edition and addenda specified in the drawing).
- ☐ ISPESL VSG / S / M Rev. 1995 Ed. 1999.



2023/6/17

Checklist completed by:	
Date:	2023.6.17

Item No	Requirement	Compliance: Yes; No; N/A	Comments/Standards used
1	General		
1.1	Pressure equipment must be designed, manufactured and checked, and if applicable equipped and installed, in such a way as to ensure its safety when put in service in accordance with manufacture's instruction.	YES	The Manufacturer has prepared Operating Instructions for the Purchaser / User containing the necessary safety information in accordance with par. 3.4 of Annex I. The Manufacturer has also considered the following aspects: ☒ inform Purchaser / User of residual hazards ☒ prevent dangers from foreseeable misuse ☒ take into account foreseeable degradation mechanisms ☒ Ensure safety throughout the intended life of the pressure equipment.
1.2	Manufacture to apply the following principles - Eliminate or reduce hazards as far as is reasonably practicable. - Apply appropriate protection measures against hazards which can not be eliminated. - Inform users of residual hazards	YES	The Manufacturer has: ☒ eliminated or reduced hazards as far as it is reasonably practical ☒ applied appropriate protection measures against hazards which cannot be eliminated ☒ informed Purchaser / User of any residual hazards and any additional measures to be taken to reduce risks during installation and / or use.
1.3	Where potential misuse is known, the pressure equipment must be designed so as to prevent danger. If that is not possible: adequate warning must be provided.	YES	☒ The Manufacturer has provided the necessary safeguards where there is potential for misuse of the pressure equipment. ☒ The Manufacturer has provided appropriate warnings in Operating Instructions regarding improper use of the equipment (see Annex I Par. 3.4).

Item No	Requirement	Compliance: Yes; No; N/A	Comments/Standards used
2	Design		
2.1 (EQ.)	General Pressure equipment must be properly designed taking all relevant factors into account in order to ensure that the equipment will be safe throughout its intended life. The design must include appropriate safety coefficients using comprehensive methods which are known to incorporate adequate safety margins against all relevant failure modes in a consistent manner.	YES	Pressure vessel Code ASME Sec VIII-1 is used to design with this ESR assessment.
2.2 (EQ.)	Design for adequate strength	YES	
2.2.1	The pressure equipment must be designed for loadings appropriate to its intended use, such as - Internal/external pressure - Ambient and operating temperatures - Static pressure and mass contents in operating and test conditions - Traffic, wind, earthquake loading - Reaction forces and moments which result from the supports, attachments, piping, etc. - Corrosion and erosion, fatigue, etc. - Decomposition of unstable fluids. Others		The following loads are considered to this job, other loads were not provided/required by the user and will be subject to user's consideration. Also refer to Operating Instructions. ☒ Internal pressure ☐ External pressure ☐ Ambient and operating temperatures ☒ Static pressure and mass contents in operating and test conditions ☐ Traffic, wind, earthquake loading ☐ Reaction forces and moments which result from the supports, attachments, piping, etc. ☒ Corrosion and erosion ☐ Fatigue ☐ Decomposition of unstable fluids.
2.2.2	Design for adequate strength must be based on: - as a general rule: a calculation method as per 2.2.3 and supplemented if necessary by an experimental design method as per 2.2.4 an experimental design method without calculation as per 2.2.4. when the $PS \cdot V$ is $< 6000 \text{ bar} \cdot L$, or $PS \cdot DN$ is $< 3000 \text{ bar}$.		Design is based on calculation method as required in ASME SEC. VIII-1.
2.2.3	Calculation method (a) Calculation method to provide sufficient safety margins (where applicable) with the requirements of Section 7. Calculation method may be carried out via: design by formula design by analysis <u>design by fracture mechanics</u>		The Manufacturer has adopted design by: ☒ formulas ☐ stress analysis ☐ fracture mechanics Sufficient safety margins are considered as required by Section 7 and ASME SEC. VIII-1.
	(b) Resistance of the pressure equipment must be established via: calculation pressure must not be less than PS, taking into account static and dynamic fluid pressures and the decomposition of unstable fluids. Calculation temperatures must allow for appropriate safety margin, taking account of all possible combinations of temperature and pressure		The Manufacturer has designed the equipment : ☒ With proper design pressure and temperature ☐ Taking in account PED requirement about decomposition of unstable fluids ☒ Considering decomposition of unstable fluid not applicable to this job
	The maximum stresses and peak stresses concentrations must be kept within safe limits, utilising appropriate values of material properties, such as yield strength, 0.2% proof stress, 1% proof stress, UTS, creep strength, plastic strain limit, impact strength, fracture toughness.		The Manufacturer has designed the equipment : ☒ By ASME SEC. VIII-1 to ensure all the stresses are within allowable safe limits. ☐ Taking in account PED requirement about the amount of creep strength / plastic strain ☒ Considering creep strength / plastic strain not applicable to this job.
	Appropriate joint factors applied to material properties depending on NDT, material joined & envisaged operating conditions, taking into account of all reasonably foreseeable degradation mechanisms (corrosion allowance, creep – design hours at specified temperatures, fatigue – design number of cycles at specified stress level)		The following joint efficiency is used based on NDT extent. ☐ 1.0 ☐ 0.85 ☒ 0.7 The following degradation mechanisms are considered: ☒ Corrosion allowance ☐ Creep ☐ Fatigue
	(c) Stability aspects: Where the calculated thickness does not allow for adequate structural stability, necessary measures (taking into account handling and transportation) to be introduced.		The design calculation has taken in account structural stability, including transport and handling.

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2.2.4	Experimental design method Test programme must be clearly defined and approved prior to testing. Test programme to include: (a) A pressure strength test (b) Where creep or fatigue exist, appropriate tests must be determined on the basis of service conditions (c) Where necessary, additional test to be conducted concerning other factors such as corrosion, external damage, etc.	N/A	The Manufacturer has ☐ adopted an experimental design method, as described in par. 2.2.4 of Annex I. The experimental program has included additional tests for : ☐ fatigue ☐ creep ☐ corrosion The program has taken in account deterioration mechanisms, where appropriate. ☒ not adopted an experimental design method
2.3.	Provisions for safe handling and operation Specified operation must preclude any reasonably foreseeable risks. Attention must be paid to: - closure & openings - dangerous discharges of pressure relief blow-off devices to prevent physical access while pressure or vacuum exists - surface temperature taking into account consideration of the intended use - decomposition of unstable fluids Access door must be equipped with an automatic or manual device enabling the user to ascertain that the opening will not present any hazard.	YES	The Manufacturer has considered these as far as applicable to this job. Manufacturing drawings show access doors below, if used. ☐ Manhole ☒ Handhole ☐ If this physical access into a vessel is a quick opening closure, has been equipped with ☐ automatic, or ☐ manual devices to enable the Purchaser / User to ascertain that the opening does not present any hazard.
2.4. (EQ.)	Means of examination (a) Pressure equipment must be design and constructed so that all necessary examinations to ensure safety can be carried out. (b) Means of determining the internal condition of the equipment must be available to ensure continued safety of the equipment (c) Other means of ensuring the safe conditions may be applied: - where it is too small for physical internal access - where opening the pressure equipment would adversely affect the inside - where the substance contained has been shown not to be harmful to the material of construction.	YES	The following means are provided for internal examination: ☐ Manhole ☒ Handhole ☐ Sight glass ☐ Process openings ☐ Substance contained has not harmful effect to the material of construction.
2.5 (EQ.)	Means of draining and ventilating Adequate means must be provided for the draining and venting of pressure equipment where necessary: - To avoid harmful effects such as water hammer, vacuum collapse, corrosion and uncontrolled chemical reactions. - To permit cleaning, inspection and maintenance in a safe manner.	YES	See manufacturing drawings and Operating Instructions Harmful effects such as water hammer, vacuum collapse and uncontrolled chemical reactions don't exist.
2.6. (EQ.)	Corrosion or other chemical attack Where necessary, adequate allowance or protection against corrosion or other chemical attack must be provided, taking into account the intended and reasonably foreseeable use.	YES	Adequate allowance is provided for corrosion and chemical attack.
2.7. (EQ.)	Wear Where sever conditions of erosion or abrasion may arise, adequate measures must be taken to: - minimise the effect by appropriate design, e.g. additional material thickness, or by the use of liners or cleaning materials. - Permit replacement of parts which are most affected. - draw attention, in the instruction referred to in 3.4, to measures necessary for continued safe use.	YES	Wear or abrasion is too light to have effect on the vessel.

Item No	Requirement	Compliance: Yes; No; N/A	Comments/Standards used
2.9.	Provisions for filling and discharge Where appropriate, pressure equipment must be so designed and provided with accessories, or made for their fitting, as to ensure safe filling and discharge. (a) on filling: - overfilling or overpressurisation having regard in particular to the filling ratio and vapour pressure at reference temperature. - Instability of the pressure equipment. (b) on discharge: the uncontrolled release of the pressurised fluid. (c) On filling or discharge: unsafe connection and disconnection.	→	The necessary accessories in accordance with the requirements of PED, to assure safe filling and discharge, prevent uncontrolled release of pressurised fluid, and prevent unsafe connection and disconnection ☐ are part of the supply ☒ are not part of the supply, but the Manufacturer has prepared Operating Instructions for the Purchaser / User containing the necessary informations
2.10.	Protection against exceeding the allowable limits of pressure equipment The pressure equipment must be fitted with, or provision made for the fitting of, suitable protective devices, unless the equipment is intended to be protected by other protective devices within an assembly. Suitable protective devices comprise: - safety accessories - where appropriate, monitoring devices.	→	The necessary indicators and alarms to comply with the requirements of PED, Annex I, par. 2.10, ☐ are part of the supply ☒ are not part of the supply, but the Manufacturer has prepared Operating Instructions for the Purchaser / User containing the necessary informations to: • design, provide and install the indicators and alarms • forbid to pressurise the equipment in lacking of above, • warn the Purchaser / User of the danger of using the equipment in lacking of above

Item No	Requirement	Compliance: Yes; No; N/A	Comments/Standards used
3	Manufacturing		
3.1.	Manufacturing procedures The manufacturer must ensure the competent execution of the provision set out at the design stage by applying the appropriate techniques and relevant procedures.	YES	The manufacturer has appropriate techniques and relevant procedures for manufacturing in their QC system.
3.1.1 (EQ.)	Preparation of the component parts Preparation of the component parts (e.g. forming and chamfering) must not give rise to defects or cracks or changes in the mechanical characteristics likely to be detrimental to the safety of the pressure equipment.	YES	Visual examination is performed after preparation of component parts. Test coupon will be prepared and tested if there is hot process affecting material properties.
3.1.2.	Permanent joining Permanent joints and adjacent zones must be free of any surface or internal defects detrimental to the safety of the equipment. The properties of the permanent joints must meet the minimum properties specified for the materials to be joined unless other relevant property values are specifically taken into account in the design calculations. Permanent joints of pressure parts must be carried out by suitably qualified personnel according to suitable operating procedures. For categories II, III and IV, operating procedures and personnel must be approved by a competent third party or NB.	NO	Visual examination and NDT method will be performed on permanent joints. Properties of permanent joints are ensured by qualified WPS and qualified welders. WPSs and welders are approved by NB.
3.1.3	Non-destructive tests NDT of permanent joints must be carried out by suitably qualified personnel. For categories III and IV the personnel must be approved by a third party organisation.	NO	
3.1.4 (EQ.)	Heat treatment Where there is a risk that the manufacturing process will change the material properties to an extent which would impair the safety of the pressure equipment, suitable heat treatment must be applied at the appropriate stage of manufacture.	N/A	Heat treatment is exempted as per design code.
3.1.5 (EQ.)	Traceability Suitable procedures must be established and maintained for identifying the material making up the components of the equipment which contribute to pressure resistance by suitable means from receipt through production, up to the final test of the manufactured pressure equipment.	YES	The manufacturer has suitable procedure for material identification, marking transfer and material traceability.
3.2	Final assessment		
3.2.1 (EQ.)	Final inspection Pressure equipment must undergo a final inspection to assess visually and by examination of the accompanying documents compliance with the requirements of PED. Tests carried out during manufacture may be taken into account. As far as is necessary, internal and external examination must be carried out.	YES	The Manufacturer has provided a quality records book to demonstrate compliance with the appropriate conformity assessment module and the Construction Code adopted.
3.2.2 (EQ.)	Proof test Final assessment of pressure equipment must include a test for the pressure containment aspect, which will normally take the form of a hydrostatic pressure test. For category I series-produced pressure equipment, pressure test may be performed on a statistical basis. Where the hydrostatic pressure test is harmful or impractical, other tests of a recognised value may be carried out.	YES	The Manufacturer has adopted : ☒ hydrostatic test the test pressure not less than values specified in par. 7.4 of Annex I and value specified in construction code, whichever is larger ☐ pneumatic test pressure not less than value specified in construction code

Item No	Requirement	Compliance: Yes; No; N/A	Comments/Standards used
3.3.	Marking and labelling	YES	
	(a) for all pressure equipment - the name and address or other means of identification of the manufacturer and, where appropriate, of his authorised representative within the EEA. - The year of manufacture - Identification of the pressure equipment according to its nature, such as type, series or batch identification and serial number. - Essential maximum/minimum allowable limit.		☒ In addition to the CE marking, the Manufacturer has provided the additional information required by PED, Annex 1, par 3.3(a) and par. 3.3(b), if appropriate (see copy of nameplate).
	(b) depending on the type of the pressure equipment, further information necessary for safe installation, operation or use and, where applicable, maintenance and periodic inspection such as: - the volume V in L - the nominal size for piping DN - the test pressure PT in bar, and date of test - safety devices set pressure in bar - output of the pressure equipment in kW - supply voltage in V (volts) - intended use - filling ratio kg/L - maximum filling mass in kg - tare mass in kg - the product group		The following information is also shown on nameplates: ☒ the volume V in L ☐ the nominal size for piping DN ☒ the test pressure PT in bar, and date of test ☐ safety devices set pressure in bar ☐ output of the pressure equipment in kW ☐ supply voltage in V (volts) ☒ intended use ☐ filling ratio kg/L ☐ maximum filling mass in kg ☒ tare mass in kg ☒ the product group
	(c) where necessary, warning fixed to the pressure equipment drawing attention to misuse which experience has shown might occur.		☐ For vessels, the Manufacturer has attached to the pressure equipment warning labels: ☐ equipment heat treated, do not weld ☐ equipment delivered filled with nitrogen at low pressure (0,5 barg) ☐ do not pressurize before installing safety devices ☐
3.4	Operating instructions	YES	Refer to Operating Instructions
	(a) when pressure equipment is placed on market, it must be accompanied, as far as relevant, with instructions for the user, containing all the necessary safety information relating to: - mounting including assembling of different pieces of pressure equipment - putting into service - use - maintenance including checks by the user. (b) instruction must cover information affixed to the pressure equipment in accordance with 3.3., with the exception of serial identification and must be accompanied, where appropriate, by the technical documents, drawings, diagrams necessary for full understanding of these instructions. If appropriate, these instructions must also refer to hazards arising from misuse in accordance with 1.3. and particular features of the design in accordance with 2.2.3.		

Item No	Requirement	Compliance: Yes; No; N/A	Comments/Standards used
4 (EQ.)	Materials	YES	
4.1.	Materials for pressurised parts must: (a) have appropriate properties for all operating conditions which are reasonably foreseeable and for all test conditions, and in particular, they should be ductile and tough (see 7.5) (b) be sufficiently chemically resistant to the fluid contained in the pressure equipment. (c) not be significantly affected by ageing. (d) be suitable for the intended processing (e) be selected in order to avoid significant undesirable effects when the various materials are put together.		☒ The Manufacturer holds responsibility for material selection for the intended service.
4.2.	(a) the pressure equipment manufacturer must define in an appropriate manner the values necessary for the design calculations referred in 2.2.3 and the essential characteristics of the materials and their treatment referred to in 4.1.		☒ The Manufacturer has used materials meeting the requirements of Annex I, par. 4.1(a) through par. 4.1(e). ☒ The Manufacturer has used steels meeting the requirements of elongation of Annex I par. 7.5.
	(b) the manufacturer must provide in his technical documentation elements relating to compliance with the materials specifications of the Directive in one of the following forms: - by using materials which comply with harmonised standards. - By using materials covered by a European approval of pressure equipment materials - By a particular material appraisal. (c) the pressure equipment in categories III and IV, particular appraisal must be performed by NB.		Selected materials conform to one or more of the following: ☐ materials that are covered by a European harmonised product standard (HS) ☐ materials that are covered by a European approval of pressure equipment materials (EAM) ☒ materials that are issued with a Particular Material Appraisal (PMA) ☒ materials that are issued with a Particular Material Appraisal (PMA) carried out by the Notified Body, for categories III and IV
4.3.	The equipment manufacturer must take appropriate measures to ensure that the material used conforms with the required specification. In particular, documentation prepared by the material manufacturer affirming compliance with a specification must be obtained for all materials. For the main pressure bearing parts in categories II, III and IV, this must take the form of a certificate of specific product control. Where a material manufacturer has an appropriate quality assurance system, certified by a competent body established within the community and having undergone a specific assessment of materials, certificates issued by the manufacturer are presumed to certify conformity with the relevant requirements.		The Manufacturer provide materials conform to the appropriate specifications and supported by appropriate certification: - for main pressure parts in categories II / III / IV: test certificates .EN 10204 types 3.2 of mechanical tests; or: 3.1 with accredited certification EN ISO 9001 or equivalent to the material manufacturer from a Body established within the EC; - for main pressure parts in category I, for attachments to pressure parts in any category, for pressure parts other than main: test certificates .EN 10204 types above specified or 2.2. - for weld filler materials in any category: ☐ AWS SFA-5.01 Level F ☒ EN 10204 type 2.2 Main pressure parts definition: see Guideline 7/6. ☒ If required by the material specification, material certificates include impact test results. irrespective of code exemption.

Item No	Requirement	Compliance: Yes; No; N/A	Comments/ Standards used
5 (EQ.)	Specific requirements: Fired or otherwise heated pressure equipment with a risk of overheating This pressure equipment includes: - steam and hot-water generators, such as fired steam and hot water boilers, superheaters and reheaters, waste-heat boilers, waste incineration boilers, electrode or immersion-type electrically heated boilers, pressure cookers, together with their accessories and where applicable their systems for treatment of feedwater and fuel supply, and - process-heating vessels for other than steam and hot water generation, such as heaters for chemical and other similar processes and pressurised food-processing equipment. This equipment must be calculated, designed and constructed so as to minimise risks of a significant loss of containment from overheating. (a) Appropriate means of protection are provided to restrict operating parameters such as heat input, heat take-off and where applicable, fluid level so as to avoid any risk of local and general overheating. (b) Sampling points are provided where required to allow evaluation of the properties of the fluid so as to avoid risks related to deposits and or corrosion. (c) Adequate provisions are made to eliminate risks of damage from deposits, (d) Means of safe removal of residual heat after shutdown are provided. (e) Steps are taken to avoid a dangerous accumulation of ignitable mixtures of combustible substances and air, or flame blowback.	N/A	

Item No	Requirement	Compliance: Yes; No; N/A	Comments/ Standards used
6 (EQ.)	Specific requirements: PIPING AS REFERRED TO IN ARTICLE 3, SECTION 1.3 Design and construction must ensure: (a) that the risk of overstressing from inadmissible free movement or excessive forces being produced, e.g. on flanges, connections, bellows or hoses, is adequately controlled by means such as support, constraint, anchoring, alignment and pre-tension; (b) that where there is a possibility of condensation occurring inside pipes for gaseous fluids, means are provided for drainage and removal of deposits from low areas to avoid damage from water hammer or corrosion; (c) that due consideration is given to the potential damage from turbulence and formation of vortices; the relevant parts of 2.7 are applicable; (d) that due consideration is given to the risk of fatigue due to vibrations in pipes; (e) that, where fluids of Group 1 are contained in the piping, appropriate means are provided to isolate 'take-off' pipes the size of which represents a significant risk; (f) that the risk of inadvertent discharge is minimized; the take-off points must be clearly marked on the permanent side, indicating the fluid contained; (g) that the position and route of underground piping is at least recorded in the technical documentation to facilitate safe maintenance, inspection or repair.	N/A	

Item No	Requirement	Compliance: Yes; No; N/A	Comments/ Standards used
7 (EQ.)	Specific quantitative requirements for certain pressure equipment Note: These provisions apply as a "general rule". Where they are not applied, the manufacturer must demonstrate that appropriate measures have been taken to achieve an "equivalent overall level of safety".	YES	
7.1	Allowable stresses		☒ See Particular Material Appraisal sheets.
7.1.1	Symbols Re/t, yield limit, indicates the value at the calculation temperature of: - the upper flow limit for a material presenting upper and lower flow limits. - The 1% proof strength for austenitic steel and non-alloyed aluminium - The 0.2% proof strength on other cases. Rm/20 indicates the minimum value of the ultimate strength at ambient temperature. Rm/t designates the ultimate strength at the calculation temperature.		

7.1.2	The permissible general membrane stress (f) for predominantly static loads and for temperature outside the range in which creep is significant, must not exceed the smaller of the following values, according to material used: - for ferritic steel., including normalised (normalised rolled) steel, [but excluding fine grained steel (especially heat treated steel)] $f = \min[2/3 Re/t, 5/12 Rm/20]$ - for austenitic steel $f = 2/3 Re/t \text{ if elongation after rupture } > 30\%$ $f = \min[5/6 Re/t, 1/3 Rm/t] \text{ if elongation } > 35\%$ - for non-ally or low alloy cast steel $f = \min[10/19 Re/t, 1/3 Rm/20]$ - for aluminium $f = 2/3 Re/t$ - for aluminium alloy [excluding precipitation hardening alloys] $f = \min[2/3 Re/t, 5/12 Rm/20]$		Allowable stresses of materials used in calculation are the lowest values as required in ASME SEC. VIII-1 and this section.
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Item No	Requirement	Compliance Yes; No; N/A	Comments/ Standards used
7.2.	Joint coefficients For welded joints, the joint coefficient must not exceed the following values - for equipment subjected to destructive and NDT which confirm that the whole series of joints show no significant defect: 1 - for equipment subject to random NDT: 0.85 - for equipment not subject to NDT other than visual inspection: 0.7		Joint efficiency below is used based on NDT extent and destructive test if applicable: ☐ for equipment subjected to destructive and NDT which confirm that the whole series of joints show no significant defect: 1 ☐ for equipment subject to random NDT: 0.85/0.8 ☒ for equipment not subject to NDT other than visual inspection: 0.7
7.3.	Pressure limiting devices, particularly for pressure vessels The momentary pressure surge referred to in 2.11.2 must be kept to 10% of the maximum allowable pressure.		Not part of supply
7.4.	Hydrostatic test pressure For pressure vessels, the hydrostatic test pressure referred to in 3.2.2 must not be less than - that corresponding to the maximum loading to which the pressure equipment may be subject in service taking into account its maximum allowable pressure and its maximum allowable temperature, multiplied by the coefficient 1.25, or - the maximum allowable pressure multiplied by the coefficient 1.43, which ever is greater.		Hydro. test at PT=20.6bar
7.5	Material characteristics Unless other values are required in accordance with other criteria that must be taken into account, a steel is considered as sufficiently ductile to satisfy 4.1 (a) if, in a tensile test carried out by a standard procedure, its elongation after rupture is no less than 14% and its bending rupture energy measured on ISO V test piece is no less than 27 J, at a temperature not greater than 20 degree C but not higher than the lowest scheduled operating temperature.		Steels used for pressure parts are tested at temperature TKV specified in Annex I par. 7.5 (or in harmonized code) and impact test energy corresponding to standard size specimens results not lesser than 27 J. ALTERNATIVELY: If so permitted by PMAs, steels used for pressure parts were not tested for impact energy if exempted by Construction Code and if this omission is allowed by criteria of guideline 7/17 and 8/6 (i. e. overall safety equivalent to PED Annex I par. 7.5 recommendations).